

$$\begin{aligned}
 1. \quad A &= \pi r^2 \\
 \frac{dA}{dr} &= 2\pi r \\
 \frac{dA}{dt} &= \frac{dA}{dr} \times \frac{dr}{dt} \\
 250 &= 2\pi(40) \times \frac{dr}{dt} \\
 \frac{dr}{dt} &= 0.995 \text{ (to 3 s.f.)} \\
 \therefore \text{Rate of increase of radius} &= \mathbf{0.995 \text{ m/h}}
 \end{aligned}$$

EXAM TIP

Apply the equation involving the rates of change: $\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$.

$$\begin{aligned}
 2. \quad (a) \quad \frac{d}{dx} \sqrt{x^2 + x + 1} &= \frac{d}{dx} (x^2 + x + 1)^{\frac{1}{2}} \\
 &= \frac{1}{2} (x^2 + x + 1)^{-\frac{1}{2}} (2x + 1) \\
 &= \frac{2x + 1}{2\sqrt{x^2 + x + 1}}
 \end{aligned}$$

$$(b) \quad \int \frac{2x + 1}{2\sqrt{x^2 + x + 1}} dx = \sqrt{x^2 + x + 1} + c$$

Multiplying both sides by 16,

$$\int \frac{16x + 8}{\sqrt{x^2 + x + 1}} dx = 16\sqrt{x^2 + x + 1} + c'$$

EXAM TIP

Use the result from part (a) that

$\int \frac{2x + 1}{2\sqrt{x^2 + x + 1}} dx = \sqrt{x^2 + x + 1} + c$. An arbitrary constant c is required for indefinite integrals.

$$\begin{aligned}
 3. \quad \text{In } \triangle PQX, \tan \alpha &= \frac{2}{x}. \\
 \text{In } \triangle PRX, \tan \beta &= \frac{3}{x}. \\
 \text{Given that } \tan(\alpha + \beta) &= 2,
 \end{aligned}$$

$$\frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} = 2$$

$$\frac{\frac{2}{x} + \frac{3}{x}}{1 - \left(\frac{2}{x}\right)\left(\frac{3}{x}\right)} = 2$$

$$\frac{5}{x} = 2 \left(1 - \frac{6}{x^2}\right)$$

$$= 2 - \frac{12}{x^2}$$

$$5x = 2x^2 - 12$$

$$2x^2 - 5x - 12 = 0$$

$$(2x + 3)(x - 4) = 0$$

$$x = -1.5 \text{ (rejected, } x > 0) \quad \text{or} \quad x = 4$$

$$\therefore x = 4$$

EXAM TIP

Find the tangent ratios for α and β in the two right-angled triangles PQX and PRX first. Then apply the Addition Formula for $\tan(\alpha + \beta)$.

$$4. \quad 2y + x = 1 \quad \text{--- (1)}$$

$$x^2 + y^2 - 2x = 4 \quad \text{--- (2)}$$

$$\text{From (1), } x = 1 - 2y \quad \text{--- (3)}$$

Substitute (3) into (2):

$$(1 - 2y)^2 + y^2 - 2(1 - 2y) = 4$$

$$1 - 4y + 4y^2 + y^2 - 2 + 4y = 4$$

$$5y^2 = 5$$

$$y^2 = 1$$

$$y = \pm 1$$

$$\text{Substitute } y = 1 \text{ into (3): } x = 1 - 2(1) = -1$$

$$\text{Substitute } y = -1 \text{ into (3): } x = 1 - 2(-1) = 3$$

$$\therefore A(-1, 1), B(3, -1)$$

$$\begin{aligned}
 \therefore \text{Midpoint of } AB &= \left(\frac{-1 + 3}{2}, \frac{1 + (-1)}{2} \right) \\
 &= \mathbf{(1, 0)}
 \end{aligned}$$

EXAM TIP

Solve the pair of equations simultaneously to obtain the coordinates of A and B . Then apply the midpoint formula.

$$\begin{aligned}
 5. \quad (a) \quad T_{r+1} &= \binom{6}{r} (2x)^{6-r} \left(-\frac{b}{x}\right)^r \\
 &= \binom{6}{r} 2^{6-r} x^{6-r} (-b)^r x^{-r} \\
 &= \binom{6}{r} 2^{6-r} (-b)^r x^{6-2r}
 \end{aligned}$$

For the term independent of x ,

$$6 - 2r = 0$$

$$2r = 6$$

$$r = 3$$

$$\binom{6}{3} 2^{6-3} (-b)^3 = -540$$

$$-160b^3 = -540$$

$$b^3 = \frac{27}{8}$$

$$b = \frac{3}{2}$$

EXAM TIP

Apply the general term of a binomial expansion $(a + b)^n$:

$$T_{r+1} = \binom{n}{r} a^{n-r} b^r.$$

- (b) For the term in x^2 ,

$$6 - 2r = 2$$

$$2r = 4$$

$$r = 2$$

$$\text{Term in } x^2 = \left(\frac{6}{2}\right)^2 \left(-\frac{3}{2}\right)^2$$

$$= 540$$

$$(1 + x^2) \left(2x - \frac{3}{2x}\right)^6 = (1 + x^2)(\dots - 540 + 540x^2 + \dots)$$

$$\text{Coefficient of } x^2 = -540 + 540$$

$$= 0$$

$\therefore$ There is no term in x^2 . (shown)

EXAM TIP

Show that the coefficient of x^2 in the expansion of

$$(1 + x^2) \left(2x - \frac{3}{2x}\right)^6 \text{ is zero.}$$

6. $\angle ABC = 90^\circ$ (rt. $\angle$ in a semicircle)

$$\angle ABX = 180^\circ - \angle ABC \text{ (adj. } \angle \text{ s on a str. line)}$$

$$= 180^\circ - 90^\circ$$

$$= 90^\circ$$

$$TA = TB \text{ (tangents from an ext. pt.)}$$

By the converse of right angle in a semicircle,

XA is a diameter of a circle, with centre at T , and passing through B .

By the converse of angle at centre = $2 \times$ angle at circumference,
 $\angle ATB = 2 \times \angle AXB$. (shown)

EXAM TIP

TA and TB are tangents from an external point T , so $TA = TB =$ radius of circle passing through X , B and A .

7. Area of base = $(2 + \sqrt{2})^2$

$$= 4 + 4\sqrt{2} + 2$$

$$= (6 + 4\sqrt{2}) \text{ cm}^2$$

$$\text{Height of cuboid} = \frac{14 + 8\sqrt{2}}{6 + 4\sqrt{2}}$$

$$= \frac{7 + 4\sqrt{2}}{3 + 2\sqrt{2}} \times \frac{3 - 2\sqrt{2}}{3 - 2\sqrt{2}}$$

$$= \frac{(7 + 4\sqrt{2})(3 - 2\sqrt{2})}{(3 + 2\sqrt{2})(3 - 2\sqrt{2})}$$

$$= \frac{21 - 14\sqrt{2} + 12\sqrt{2} - 16}{9 - 8}$$

$$= (5 - 2\sqrt{2}) \text{ cm}$$

$$\text{Total surface area} = 2(6 + 4\sqrt{2}) + 4(2 + \sqrt{2})(5 - 2\sqrt{2})$$

$$= 12 + 8\sqrt{2} + 4(10 - 4\sqrt{2} + 5\sqrt{2} - 4)$$

$$= 12 + 8\sqrt{2} + 40 - 16\sqrt{2} + 20\sqrt{2} - 16$$

$$= (36 + 12\sqrt{2}) \text{ cm}^2$$

EXAM TIP

$$\text{Height of cuboid} = \frac{\text{Volume of cuboid}}{\text{Base area}}$$

To find the height of the cuboid, rationalise the denominator by multiplying the numerator and the denominator by its conjugate.

8. (a) Total surface area = $2x^2 + 4xy$

$$2x^2 + 4xy = 600$$

$$4xy = 600 - 2x^2$$

$$y = \frac{600 - 2x^2}{4x}$$

$$= \frac{300 - x^2}{2x}$$

$$\therefore \text{Volume of box, } V = x^2 \left(\frac{300 - x^2}{2x} \right)$$

$$= 150x - \frac{x^3}{2} \text{ (shown)}$$

$$(b) V = 150x - \frac{x^3}{2} \quad \text{--- (1)}$$

$$\frac{dV}{dx} = 150 - \frac{3}{2}x^2$$

$$\text{When } V \text{ is stationary, } \frac{dV}{dx} = 0.$$

$$150 - \frac{3}{2}x^2 = 0$$

$$\frac{3}{2}x^2 = 150$$

$$x^2 = 100$$

$$x = 10 \quad \text{or} \quad x = -10 \text{ (rejected, } x > 0)$$

Substitute $x = 10$ into (1):

$$V = 150(10) - \frac{10^3}{2}$$

$$= 1000$$

$$\frac{d^2V}{dx^2} = -3x$$

$$\text{When } x = 10,$$

$$\frac{d^2V}{dx^2} = -30 < 0$$

$\therefore V = 1000$ is a maximum.

EXAM TIP

For a stationary value of V , $\frac{dV}{dx} = 0$. If $\frac{d^2V}{dx^2} > 0$, then V is a minimum. If $\frac{d^2V}{dx^2} < 0$, then V is a maximum.

9. (a) (i) $2x^2 - 6x + 7 = 2(x^2 - 3x) + 7$

$$= 2 \left[\left(x - \frac{3}{2} \right)^2 - \frac{9}{4} \right] + 7$$

$$= 2 \left(x - \frac{3}{2} \right)^2 + \frac{5}{2}$$

$$(ii) \quad 2x^2 - 6x + 7 = k$$

$$2x^2 - 6x + 7 - k = 0$$

Since the equation does not have two distinct roots,

$$\text{Discriminant} \leq 0$$

$$(-6)^2 - 4(2)(7 - k) \leq 0$$

$$36 - 56 + 8k \leq 0$$

$$8k \leq 20$$

$$k \leq \frac{5}{2}$$

EXAM TIP

Since the equation does not have two distinct roots, then it can have two real and equal roots, or no real roots.

$$(b) \text{ x-coordinate of maximum point} = \frac{2+4}{2} = 3$$

Coordinates of maximum point: (3, 2)

$$\therefore q = 3, r = 2$$

Substitute $x = 2, y = 0$ into $y = p(x-3)^2 + 2$:

$$0 = p(1) + 2$$

$$p = -2$$

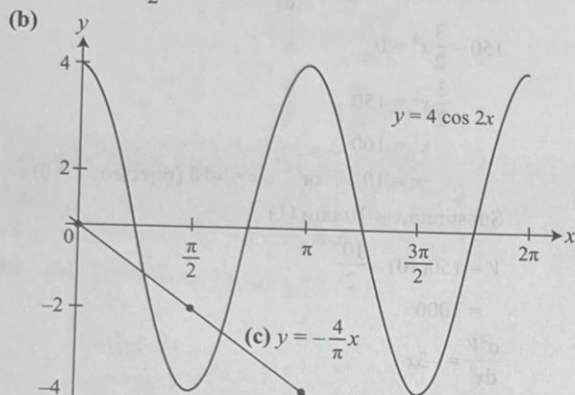
$$\therefore p = -2, q = 3, r = 2$$

EXAM TIP

A quadratic curve is symmetrical about the vertical line passing through its turning point.

10. (a) Amplitude = 4

$$\text{Period} = \frac{2\pi}{2} = \pi$$



(c) $\pi \cos 2x + x = 0$

$$4\pi \cos 2x + 4x = 0$$

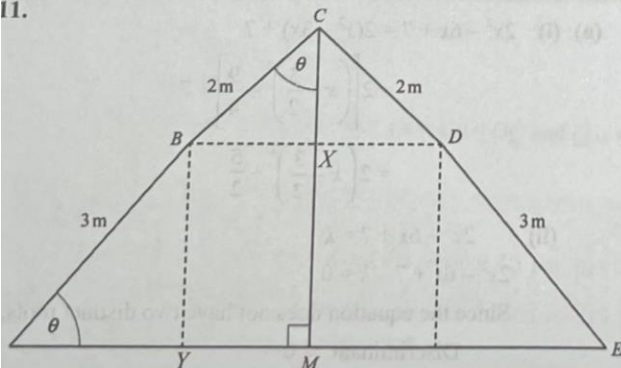
$$4 \cos 2x + \frac{4}{\pi}x = 0$$

$$4 \cos 2x = -\frac{4}{\pi}x$$

$$\text{Draw } y = -\frac{4}{\pi}x.$$

From the graph, there are 2 solutions.

11.



(a) Consider $\triangle CBX$.

$$\cos \theta = \frac{CX}{2}$$

$$CX = 2 \cos \theta$$

Consider $\triangle ABY$.

$$\sin \theta = \frac{BY}{3}$$

$$BY = 3 \sin \theta$$

$$\therefore CM = CX + XM$$

$$= CX + BY$$

$$= 2 \cos \theta + 3 \sin \theta \text{ (shown)}$$

(b) $2 \cos \theta + 3 \sin \theta = R \cos (\theta - \alpha)$

$$R = \sqrt{2^2 + 3^2}$$

$$= \sqrt{13}$$

$$\tan \alpha = \frac{3}{2}$$

$$\alpha = 56.3^\circ \text{ (to 1 d.p.)}$$

$$\therefore 2 \cos \theta + 3 \sin \theta = \sqrt{13} \cos (\theta - 56.3^\circ)$$

$$\therefore \text{Maximum possible length of } CM = \sqrt{13} \text{ m}$$

EXAM TIP

Apply the R-Formula: $a \cos \theta + b \sin \theta = R \cos (\theta - \alpha)$, where

$R = \sqrt{a^2 + b^2}$, $\tan \alpha = \frac{b}{a}$ and α is acute.

Maximum value of $R \cos (\theta - \alpha) = R$.

(c) $2 \cos \theta + 3 \sin \theta = 3.5$

$$\sqrt{13} \cos (\theta - 56.310^\circ) = 3.5$$

$$\cos (\theta - 56.310^\circ) = 0.97073 \text{ (to 5 s.f.)}$$

Basic angle = 13.898° (to 3 d.p.)

$$\theta - 56.310^\circ = -13.898^\circ, 13.898^\circ \text{ (to 3 d.p.)}$$

$$\theta = 42.4^\circ, 70.2^\circ \text{ (to 1 d.p.)}$$

If $\theta = 42.4^\circ$, $\angle ABC > 180^\circ$ (reflex angle).

If $\theta = 70.2^\circ$, $\angle ABC < 180^\circ$ (obtuse angle).

$$\therefore \theta = 70.2^\circ$$

12. (a) Let $\frac{3x-5}{x^2(x-1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-1}$.

$$3x - 5 = Ax(x-1) + B(x-1) + Cx^2$$

$$\text{Let } x = 0: -5 = -B$$

$$B = 5$$

$$\text{Let } x = 1: -2 = C$$

Equating coefficients of x^2 ,

$$0 = A - 2$$

$$A = 2$$

$$\therefore \frac{3x-5}{x^2(x-1)} = \frac{2}{x} + \frac{5}{x^2} - \frac{2}{x-1}$$

EXAM TIP

Let $x = 0$ to eliminate the terms containing A and C . Then use substitution or equate coefficients to find the values of A and C .

$$\begin{aligned}
 \text{(b)} \quad \int_2^3 \frac{3x-5}{x^2(x-1)} dx &= \int_2^3 \left(\frac{2}{x} + \frac{5}{x^2} - \frac{2}{x-1} \right) dx \\
 &= \int_2^3 \left(\frac{2}{x} + 5x^{-2} - \frac{2}{x-1} \right) dx \\
 &= \left[2 \ln x + \frac{5x^{-1}}{-1} - 2 \ln(x-1) \right]_2^3 \\
 &= \left[2 \ln \frac{x}{x-1} - \frac{5}{x} \right]_2^3 \\
 &= 2 \ln \frac{3}{2} - \frac{5}{3} - 2 \ln 2 + \frac{5}{2} \\
 &= \frac{5}{6} - 2 \left(\ln 2 - \ln \frac{3}{2} \right) \\
 &= \frac{5}{6} - 2 \ln \frac{4}{3} \\
 &= \frac{5}{6} - \ln \left(\frac{4}{3} \right)^2 \\
 &= \frac{5}{6} - \ln \frac{16}{9} \text{ (shown)}
 \end{aligned}$$

EXAM TIP

Recall: $\int \frac{1}{x} dx = \ln x + c$ and $\int \frac{1}{x^2} dx = \frac{x^{-1}}{-1} + c = -\frac{1}{x} + c$.

13. (a) Substitute $x = 0$ into $y = (x-1)^4$:

$$y = 1$$

$$\therefore Q(0, 1)$$

$$y = (x-1)^4$$

$$\frac{dy}{dx} = 4(x-1)^3(1)$$

$$= 4(x-1)^3$$

$$\text{When } x = 2, \frac{dy}{dx} = 4.$$

$$\text{Gradient of } PS = -\frac{1}{4}$$

Let the coordinates of S be $(0, y_S)$.

$$\frac{y_S - 1}{0 - 2} = -\frac{1}{4}$$

$$y_S - 1 = \frac{1}{2}$$

$$y_S = 1.5$$

Since $OQ = 1$ and $QS = 0.5$, then $QS < OQ$ and Q is nearer to S .

EXAM TIP

Alternatively, find the equation of the normal PS and use it to find the coordinates of S .

- (b) Substitute $y = 0$ into $y = (x-1)^4$:

$$x = 1$$

Let the coordinates of R be $(x_R, 0)$.

$$\frac{1-0}{2-x_R} = 4$$

$$1 = 8 - 4x_R$$

$$4x_R = 7$$

$$x_R = \frac{7}{4}$$

$$\begin{aligned}
 \text{Area of shaded region} &= \int_1^2 (x-1)^4 dx - \frac{1}{2} \left(\frac{1}{4} \right) (1) \\
 &= \left[\frac{(x-1)^5}{5(1)} \right]_1^2 - \frac{1}{8} \\
 &= \left[\frac{(x-1)^5}{5} \right]_1^2 - \frac{1}{8} \\
 &= \frac{1}{5} - \frac{1}{8} \\
 &= \frac{3}{40} \text{ unit}^2
 \end{aligned}$$

EXAM TIP

Area of shaded region = Area between curve and x -axis from $x = 1$ to $x = 2$ - area of unshaded region between PR and x -axis.

Paper 2

October/November 2024

1. $V = 1800e^{-kt}$

When $t = 0$, $V = 1800$.

When $t = 12$, $V = 1000$.

$$1000 = 1800e^{-12k}$$

$$e^{-12k} = \frac{5}{9}$$

$$-12k = \ln \frac{5}{9}$$

$$k = -\frac{1}{12} \ln \frac{5}{9}$$

$$= 0.048\ 982 \text{ (to 5 s.f.)}$$

When $V = 25\% \times 1800 = 450$,

$$450 = 1800e^{-0.048\ 982t}$$

$$e^{-0.048\ 982t} = 0.25$$

$$-0.048\ 982t = \ln 0.25$$

$$t = \frac{\ln 0.25}{-0.048\ 982}$$

$$= 28.3 \text{ (to 3 s.f.)}$$

$$\therefore t = 28.3$$

EXAM TIP

Use the fact that $t = 12$ and $V = 1000$ to find the value of k first.

2. $y = kx$ — (1)

$$y = x^2 + 3kx + 2 - k$$
 — (2)

Substitute (2) into (1):

$$x^2 + 3kx + 2 - k = kx$$

$$x^2 + 2kx + 2 - k = 0$$

Since the line intersects the curve at two distinct points,

$$\text{Discriminant} > 0$$

$$(2k)^2 - 4(1)(2-k) > 0$$

$$4k^2 - 8 + 4k > 0$$

$$k^2 + k - 2 > 0$$

$$(k+2)(k-1) > 0$$

$$\therefore k < -2 \text{ or } k > 1$$

EXAM TIP

If a line intersects a curve at two distinct points, then the discriminant > 0 .

3. $\log_2 x - \log_4 (x+3) = 3 \log_8 2$

$$\log_2 x - \frac{\log_2 (x+3)}{\log_2 4} = \log_8 2^3$$

$$\log_2 x - \frac{\log_2 (x+3)}{\log_2 2^2} = \log_8 8$$

$$\log_2 x - \frac{1}{2} \log_2 (x+3) = 1$$

$$2 \log_2 x - \log_2 (x+3) = 2$$

$$\log_2 x^2 - \log_2 (x+3) = 2$$

$$\log_2 \frac{x^2}{x+3} = 2$$

$$\frac{x^2}{x+3} = 4$$

$$x^2 = 4x + 12$$

$$x^2 - 4x - 12 = 0$$

$$(x-6)(x+2) = 0$$

$$x = 6 \quad \text{or} \quad x = -2 \text{ (rejected, } x > 0)$$

$$\therefore x = 6$$

EXAM TIP

Apply the Change of Base Formula: $\log_a b = \frac{\log_c b}{\log_c a}$.

4. (a) (i) $y = \frac{4x}{5-3x}$

$$\frac{dy}{dx} = \frac{(5-3x) \times 4 - 4x \times (-3)}{(5-3x)^2}$$

$$= \frac{20 - 12x + 12x}{(5-3x)^2}$$

$$= \frac{20}{(5-3x)^2}$$

$$\text{When } x = 1, \frac{dy}{dx} = 5.$$

$\therefore$ Gradient of curve at $x = 1$ is 5

EXAM TIP

Apply the Quotient Rule: $\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$.

(ii) Since $(5-3x)^2 > 0$ for all real values of x , $x \neq \frac{5}{3}$,
then $\frac{20}{(5-3x)^2} > 0$.

Hence, $\frac{dy}{dx} \neq 0$ and the curve has no stationary points.

(b) $f'(x) = \frac{4}{\cos^2 x}$
 $= 4 \sec^2 x$

$$f(x) = \int 4 \sec^2 x \, dx$$

$$= 4 \tan x + c$$

$$\text{Since } f\left(\frac{\pi}{4}\right) = 1,$$

$$1 = 4 \tan \frac{\pi}{4} + c$$

$$= 4 + c$$

$$c = -3$$

$$\therefore f(x) = 4 \tan x - 3$$

EXAM TIP

Recall: $\frac{1}{\cos^2 x} = \sec^2 x$ and $\int \sec^2 x \, dx = \tan x + c$.

5. (a) $y = e^x + e^{-x}$

$$\frac{dy}{dx} = e^x - e^{-x}$$

$$\text{When } \frac{dy}{dx} = 0,$$

$$e^x - e^{-x} = 0$$

$$e^x = e^{-x}$$

$$= \frac{1}{e^x}$$

$$e^{2x} = 1$$

$$2x = 0$$

$$x = 0$$

$\therefore$ The curve has a stationary point at $x = 0$. (shown)

(b) $e^x + e^{-x} = 2.9$

$$e^x + \frac{1}{e^x} = 2.9$$

$$e^{2x} - 2.9e^x + 1 = 0$$

$$e^x = \frac{-(-2.9) \pm \sqrt{(-2.9)^2 - 4(1)(1)}}{2(1)}$$

$$= \frac{2.9 \pm \sqrt{4.41}}{2}$$

$$= \frac{2.9 \pm 2.1}{2}$$

$$= 2.5 \text{ or } 0.4$$

$$x = \ln 2.5 \text{ or } x = \ln 0.4$$

EXAM TIP

Alternatively, form a quadratic equation in e^x by replacing e^x with another variable, u .

6. (a) Let $f(x) = x^3 + x^2 + ax + 2b$

$$g(x) = x^3 + 4x^2 - ax + b$$

By the Factor Theorem,

$$f(-3) = 0 \text{ and } g(-3) = 0.$$

$$-27 + 9 - 3a + 2b = 0$$

$$3a - 2b = -18 \quad \text{--- (1)}$$

$$-27 + 36 + 3a + b = 0$$

$$3a + b = -9 \quad \text{--- (2)}$$

$$(2) - (1): 3b = 9$$

$$b = 3$$

Substitute $b = 3$ into (2):

$$3a + 3 = -9$$

$$3a = -12$$

$$a = -4$$

$$\therefore a = -4, b = 3$$

EXAM TIP

If $f(x)$ is exactly divisible by $x + 3$, then $x = -3$ is a solution.

(b) Let $f(x) = x^3 + 2x^2 - 11x + 6$.

$$f(2) = 2^3 + 2(2)^2 - 11(2) + 6$$

$$= 0$$

$\therefore x - 2$ is a factor of $f(x)$.

$$f(x) = (x - 2)(x^2 + bx - 3)$$

Equating coefficients of x^2 ,

$$2 = b - 2$$

$$b = 4$$

$$f(x) = (x - 2)(x^2 + 4x - 3)$$

$$\text{When } x^3 + 2x^2 - 11x + 6 = 0,$$

$$(x - 2)(x^2 + 4x - 3) = 0$$

$$x = 2 \quad \text{or} \quad x = \frac{-4 \pm \sqrt{4^2 - 4(1)(-3)}}{2(1)}$$

$$= \frac{-4 \pm \sqrt{28}}{2(1)}$$

$$= \frac{-4 \pm 2\sqrt{7}}{2(1)}$$

$$= -2 \pm \sqrt{7}$$

$$\therefore x = 2, x = -2 + \sqrt{7} \text{ or } x = -2 - \sqrt{7}$$

EXAM TIP

Find a linear factor of $f(x)$ first. Then use long division or equate coefficients to find the quadratic factor of $f(x)$.

7. (a) Centre of circle = midpoint of PQ

$$= \left(\frac{3 + (-5)}{2}, \frac{1 + 7}{2} \right)$$

$$= (-1, 4)$$

$$PQ = \sqrt{[3 - (-5)]^2 + (1 - 7)^2}$$

$$= \sqrt{64 + 36}$$

$$= \sqrt{100}$$

$$= 10 \text{ units}$$

$$\text{Radius of circle} = 5 \text{ units}$$

EXAM TIP

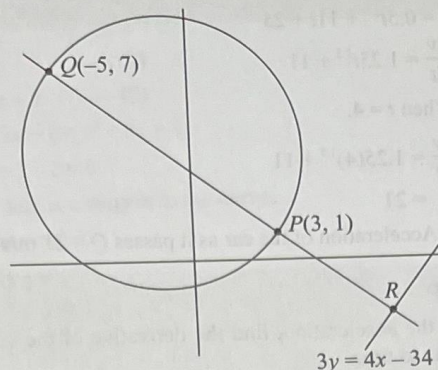
The midpoint of the diameter is the centre of the circle.

(b) Equation of circle: $(x + 1)^2 + (y - 4)^2 = 5^2$

$$x^2 + 2x + 1 + y^2 - 8y + 16 = 25$$

$$x^2 + y^2 + 2x - 8y - 8 = 0$$

(c)



$$\begin{aligned} \text{Gradient of } PQ &= \frac{7 - 1}{-5 - 3} \\ &= -\frac{3}{4} \end{aligned}$$

$$\text{Gradient of } 3y = 4x - 34 \text{ is } \frac{4}{3}$$

Since gradient of $PQ \times$ gradient of $3y = 4x - 34$ is -1 , then PQ is perpendicular to $3y = 4x - 34$.

$\therefore PR$ is the shortest distance from the line to the circle, and R lies on QP extended.

EXAM TIP

Recall that if $m_1 \times m_2 = -1$, then the two lines are perpendicular to each other.

(d) Equation of PQ : $y - 1 = -\frac{3}{4}(x - 3)$

$$= -\frac{3}{4}x + \frac{9}{4}$$

$$y = -\frac{3}{4}x + \frac{13}{4} \quad \text{--- (1)}$$

Substitute (1) into $3y = 4x - 34$:

$$\begin{aligned} 3\left(-\frac{3}{4}x + \frac{13}{4}\right) &= 4x - 34 \\ -9x + 39 &= 16x - 136 \\ 25x &= 175 \\ x &= 7 \end{aligned}$$

Substitute $x = 7$ into (1):

$$\begin{aligned} y &= -\frac{3}{4}(7) + \frac{13}{4} \\ &= -2 \end{aligned}$$

$$\therefore R(7, -2)$$

8. (a) $v = At^{2.5} + 11t + C$

$$\text{When } t = 0, v = 25,$$

$$25 = A(0) + 11(0) + C$$

$$C = 25$$

$$v = At^{2.5} + 11t + 25$$

$$\text{When } t = 4, v = 85,$$

$$85 = A(4)^{2.5} + 11(4) + 25$$

$$= 32A + 44 + 25$$

$$32A = 16$$

$$A = 0.5$$

$$(b) v = 0.5t^{2.5} + 11t + 25$$

$$\frac{dv}{dt} = 1.25t^{1.5} + 11$$

When $t = 4$,

$$\begin{aligned}\frac{dv}{dt} &= 1.25(4)^{1.5} + 11 \\ &= 21\end{aligned}$$

$\therefore$ Acceleration of the car as it passes $Q = 21 \text{ m/s}^2$

EXAM TIP

To obtain the acceleration, find the derivative of the velocity with respect to time.

$$(c) v = 0.5t^{2.5} + 11t + 25$$

$$s = \int_0^4 (0.5t^{2.5} + 11t + 25) dt$$

$$= \left[\frac{0.5t^{3.5}}{3.5} + \frac{11t^2}{2} + 25t \right]_0^4$$

$$= \frac{1444}{7} - 0$$

$$= 206 \text{ m (to the nearest metre)}$$

EXAM TIP

Apply: distance between P and $Q = \int_0^4 v dt$.

$$9. (a) \text{ LHS} = \tan x + \cot x$$

$$= \frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}$$

$$= \frac{\sin^2 x + \cos^2 x}{\sin x \cos x}$$

$$= \frac{1}{\sin x \cos x}$$

$$= \frac{2}{2 \sin x \cos x}$$

$$= \frac{2}{\sin 2x}$$

$$= 2 \operatorname{cosec} 2x$$

$$= \text{RHS (proven)}$$

EXAM TIP

Apply the formulae: $\sin^2 x + \cos^2 x = 1$ and $\sin 2x = 2 \sin x \cos x$.

$$(b) \tan x + \cot x = 3$$

$$2 \operatorname{cosec} 2x = 3$$

$$\operatorname{cosec} 2x = \frac{3}{2}$$

$$\sin 2x = \frac{2}{3}$$

Basic angle $= 0.729 73$ (to 5 s.f.)

$$2x = 0.729 73, 2.4119, 7.0129, 8.6951 \text{ (to 5 s.f.)}$$

$$x = 0.365, 1.21, 3.51, 4.35 \text{ (to 3 s.f.)}$$

EXAM TIP

Since $0 \leq x \leq 2\pi$, then $0 \leq 2x \leq 4\pi$.

$$(c) \tan x + \cot x = k$$

$$2 \operatorname{cosec} 2x = k$$

$$\operatorname{cosec} 2x = \frac{k}{2}$$

$$\sin 2x = \frac{2}{k}$$

Since $-1 \leq \sin 2x \leq 1$, then $\sin 2x = \frac{2}{k}$ has no solutions when $\frac{2}{k} < -1$ or $\frac{2}{k} > 1$.

$\therefore$ The equation has no solutions in the range $-2 < k < 2$.

EXAM TIP

For $\sin 2x = \frac{2}{k}$ to have no solutions, $\frac{2}{k} < -1$ or $\frac{2}{k} > 1$.

Multiply both sides of each inequality by k^2 ,

$$2k < -k^2 \text{ or } 2k > k^2$$

$$k^2 + 2k < 0 \quad k^2 - 2k < 0$$

$$k(k+2) < 0 \quad k(k-2) < 0$$

$$-2 < k < 0 \quad 0 < k < 2$$

$$\therefore -2 < k < 2$$

$$10. (a) (i) R = Av^k$$

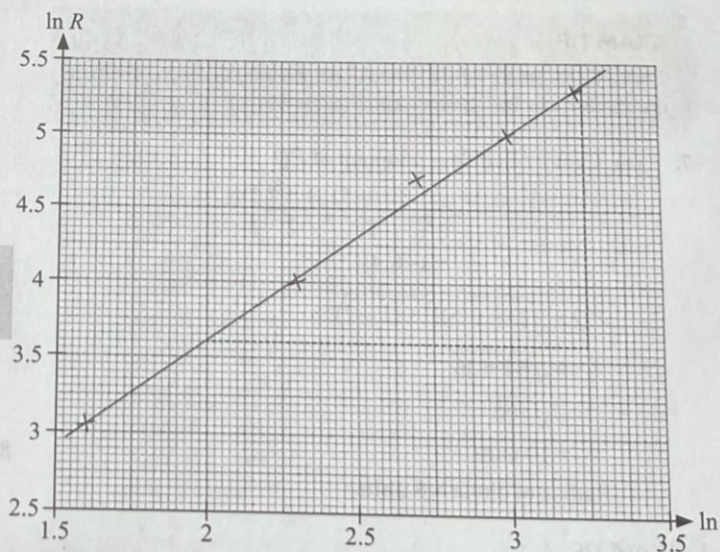
$$\ln R = \ln Av^k$$

$$= \ln A + \ln v^k$$

$$= k \ln v + \ln A$$

Plot $\ln R$ against $\ln v$.

$\ln v$	1.61	2.30	2.71	3.00	3.22
$\ln R$	3.04	4.01	4.70	4.98	5.30



(ii) From the graph, the incorrect reading is $R = 110$.

The correct reading is $R = 97$. (to 2 s.f.)

(iii) $k = \text{gradient}$

$$= \frac{5.35 - 3.6}{3.25 - 2}$$

$$= 1.4$$

EXAM TIP

The value of k can be estimated by finding the gradient of the graph.

$$(b) y = ax + \frac{b}{x}$$

$$\frac{y}{x} = \frac{b}{x^2} + a$$

$$\frac{y}{x} = b\left(\frac{1}{x^2}\right) + a$$

Plot $\frac{y}{x}$ on the vertical axis against $\frac{1}{x^2}$ on the horizontal axis.

Gradient of line = b

Vertical intercept ($\frac{y}{x}$ -intercept) of line = a

EXAM TIP

Alternatively, from $y = ax + \frac{b}{x}$, we have $xy = ax^2 + b$.

Plot xy against x^2 . The gradient is a , and the vertical intercept is b .

11. (a) $e^{-2x} > 0$ for all real values of x .

$\sin x > 0$ for $0 < x < \pi$.

$\therefore y = e^{-2x} \sin x$ lies completely above the x -axis for $0 < x < \pi$.

EXAM TIP

The graph of $y = e^{-2x}$ lies completely above the x -axis for all real values of x .

The graph of $y = \sin x$ lies above the x -axis for $0 < x < \pi$.

$$(b) y = e^{-2x} \sin x$$

$$\frac{dy}{dx} = e^{-2x} \cos x + \sin x (-2e^{-2x})$$

$$= e^{-2x} \cos x - 2e^{-2x} \sin x$$

$$= e^{-2x}(\cos x - 2 \sin x)$$

When $\frac{dy}{dx} = 0$,

$$e^{-2x}(\cos x - 2 \sin x) = 0$$

$$e^{-2x} = 0 \text{ (no solutions)} \quad \text{or} \quad \cos x - 2 \sin x = 0$$

$$2 \sin x = \cos x$$

$$\tan x = \frac{1}{2}$$

$$x = \tan^{-1} \frac{1}{2}$$

(shown)

EXAM TIP

Recall: $\frac{d}{dx}(e^{ax}) = ae^{ax}$ and $\frac{d}{dx}(\sin x) = \cos x$.

$$(c) \frac{d^2y}{dx^2} = e^{-2x}(-\sin x - 2 \cos x) + (\cos x - 2 \sin x)(-2e^{-2x})$$

$$= e^{-2x}(-\sin x - 2 \cos x) - 2e^{-2x}(\cos x - 2 \sin x)$$

$$= e^{-2x}(-\sin x - 2 \cos x - 2 \cos x + 4 \sin x)$$

$$= e^{-2x}(3 \sin x - 4 \cos x)$$

$$\frac{d^2y}{dx^2} + 4 \frac{dy}{dx} + 5y$$

$$= e^{-2x}(3 \sin x - 4 \cos x) + 4e^{-2x}(\cos x - 2 \sin x)$$

$$+ 5(e^{-2x} \sin x)$$

$$= e^{-2x}(3 \sin x - 4 \cos x + 4 \cos x - 8 \sin x + 5 \sin x)$$

$$= 0 \text{ (shown)}$$

EXAM TIP

Recall: $\frac{d}{dx}(\cos x) = -\sin x$.